ACE
Engineering College

An Autonomous Institution

Ankushapur (V), Ghatkesar (M), Medchal Dist. – 501 301

Assignment 1(A): Java Foundations & OOP Principles

Last Date for Submission: 21-03-2026 **Course**: IT405PC Java Programming

Total Marks: 2.5

Unit I (Basics, Classes, and the four pillars of OOP)

1. Explain the features that make Java "**Platform Independent.**" Write a program to demonstrate the use of ternary operators and switch-case statements to determine if a given year is a leap year.

2. Describe the role of Constructors and the **this** keyword. Create a **Student** class with private attributes, a parameterized constructor, and **getter/setter** methods to demonstrate Encapsulation.

3. Distinguish between Method Overloading and Method Overriding. Write a program where a **SocialMediaAccount** base class is inherited by **InstagramAccount**, using the **super** keyword to invoke the parent constructor.

4. Compare Abstract Classes with Interfaces. Provide a code example implementing an interface **PaymentGateway** with a method **processTransaction()** that is defined by two different classes: **UPI** and **CreditCard**.

Note:

- All assignments must be submitted on or before the mentioned date; late submissions will not be accepted, and students who fail to submit will be awarded 0 marks.

Presentation Guidelines:

- Use A4 sheets. Write your Name, Roll Number, Section, and Department clearly on the first page.
- Maintain logical flow and neat presentation throughout.

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Assignment 1(B): Errors, Files and Package Management

Last Date for Submission: 21-03-2026 **Course**: IT405PC Java Programming

Total Marks: 2.5

Unit II & Unit III (Exceptions, I/O, and Packages)

1. Explain the Java Exception Hierarchy. Write a program that uses **try-catch-finally** blocks and includes a Custom Exception called **InvalidAgeException** which is thrown if a user enters an age less than 18.

2. Discuss the difference between Byte Streams and Character Streams. Write a Java program using **FileOutputStream** to write a string to a file and **FileInputStream** to read it back and print it to the console.

3. Explain the importance of the **CLASSPATH** and Access Specifiers in package management. Create a package named **com.math.operations** containing a class with a method to calculate the factorial of a number, then import it into a different project.

4. What is the purpose of the **RandomAccessFile** class? Write a snippet that demonstrates how to move the file pointer to a specific position to update a specific byte of data in a text file.

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